SafeHer : Women Safety Platform

(An AI-powered platform designed to empower and protect women).

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY PUNE

Leading Technology. Leveraging Technology.

Prepared By:

Sawan Tanwar (112316044)
Sonu Krishniya (112315187)
Piyush Singhal (112316035)
Adarsh Beriwala (112316002)

Supervised by:

Dr. Sushant Kumar

Introduction

- **What's the problem?**

1 in 3 women worldwide experience physical violence. This isn't a distant statistic; it's a reality for millions. Personal safety, especially for women, is a critical concern in today's world. In emergency situations, every second counts, and the ability to alert loved ones and authorities quickly is vital.

- **What's the need?**

A rapid, reliable, and intelligent system that can:

1. Be triggered silently and hands-free without drawing attention.
2. Provide real-time, actionable information (like live location) and some visual/audio evidence.
3. Send an emergency alert message to your trusted ones for them to act quickly.
4. Function as a proactive safety companion, not just a reactive tool.

Literature Study - Research Papers

AUTHORS	Paper Title	Published in	Year	Findings
P. Premi, K.S. Savita, N. Millatina	FRNDY: A Women's Safety App	6th International Conference On Computing, Communication, Control And Automation (ICCUBEA)	2022	FRNDY successfully developed a free, bug-free Android app with GPS tracking, SOS alerts, and emergency contact features as a cost-effective safety solution.
Dr. Sridhar Mandapati, Sravya Pamidi, Sriharitha Ambati	A mobile based women safety application (I Safe Apps)	IOSR Journal of Computer Engineering (IOSR-JCE)	2015	I-Safe successfully developed a one-touch emergency alert system with location sharing, fake calls, and automatic guardian notification features.
Dhruv Chand, Sunil Nayak, Karthik S. Bhat, Shivani Parikh, Yuvraj Singh, Amita Ajith Kamath	A mobile application for Women's Safety: WoSApp	TENCON IEEE region 10 conference	2015	WoSApp can be activated discreetly by shaking the phone or by pressing a "PANIC" button, which sends the user's location and pre-selected emergency contacts to the police.
Aysha Sanam Ummer, Bettina Mariam Biju, Joanna Merine Noble, Sreya K U, Divya K S	HER SHIELD: Speech Emotion Recognition for Women’s Safety	2nd International Conference on Trends in Engineering Systems and Technologies (ICTEST)	2025	The system aims to automatically detect emotions like fear or distress to trigger an emergency alert, providing a hands-free method to call for help in dangerous situations.
Kalyani Ghuge, Samruddhi Khillari, Aarya Sadawrate, Rutuja Varpe, Aman Shaikh, Rajvardhan Desai	FEARLESS: Women Safety Software	IEEE International Conference on Computing Technologies (ICOCT)	2025	Integrated real-time tracking, safe route prediction, triply-triggered SOS from the power button, and a backend with location, crime, and traffic data.

Basepaper Overview

FEARLESS: Women Safety Software

Authors: Prof. Kalyani Ghuge, Rutuja Varpe, Samruddhi Khillari, Aman Shaikh, Aarya Sadawrate, Rajvardhan Desai

Published in : IEEE International Conference on Computing Technologies (ICOCT), 2025

Research Objective

To design and develop a mobile application that enhances women's safety through real-time tracking, safest-route guidance, and instant emergency alerts. The system integrates crime data, traffic updates, and local news to help women travel safely and react swiftly during emergencies, ensuring proactive and reliable protection.

Area	Summary
Dataset	Pune crime data with safety scores per area.
Feature Extraction	Combined crime, traffic, and local safety data.
System Design	Firebase + Google Maps + Twilio integrated system.
Accuracy	98% SOS success, <500 ms time required to compute route , 85% safer travel.
Innovation	Triple power-button SOS & data-driven safe navigation.

Basepaper Overview

(Dataset and Safety Score Methodology)

i). Dataset Characteristics:

The system utilizes crime pattern data from Pune City containing reported crime numbers across various police stations, including police station names, crime types, monthly crime counts (January-September), and geographical coordinates.

ii.) Safety Score Calculation Formula:

The safety score quantifies relative safety levels across locations using min-max normalization:

$$\boxed{\text{Safety\_Score\_Station}_i = 1 - \left(\frac{\text{Total\_Crime}_i - \min(\text{Total\_Crime})}{\text{Max}(\text{Total\_Crime}) - \text{Min}(\text{Total\_Crime})}\right)}$$

Where:
Safety_Score_Station_i = safety score for station i (0 = least safe, 1 = safest)
Total_Crime_i = total crimes reported at station i
Values are normalized across all stations for comparative analysis

iii.) Data Processing Pipeline:

Raw Crime Data → Cleaning & Aggregation → Coordinate Parsing → Safety Score Calculation → JSON Conversion → Geospatial Analysis

Basepaper Overview

(Safety Route Finding and Heatmap Visualization)

i). <u>Safest Route Navigation Algorithm</u>:

The system evaluates multiple potential routes using a weighted safety scoring mechanism:

$$\boxed{\text{Safety\_Score\_Route (R)} = \frac{\sum_{j=1}^{n}(S_j \times D_j)}{\sum_{j=1}^{n} D_j}}$$

Where:
s_i = safety score at route segment i
d_i = distance of segment i
n = total segments in route R

ii.) <u>Safest route Selection Methodology</u>:

- FEARLESS determines the safest navigation path by integrating crime data with location analysis.
- Users enter start and end points, which are converted into precise geographic coordinates.
- The system uses Google Directions API to fetch multiple route options.
- Only routes with at least 90% of their path within Pune's boundaries are considered.
- Each route is divided into consecutive segments for detailed analysis.
- At each segment's midpoint, the system averages safety scores of nearby crime data points within a 500-meter radius.
- Segment safety scores are weighted according to segment length to prioritize longer safe segments.
- The route with the highest overall weighted safety score is selected and shown to the user.

Basepaper Overview

(Safety Route Finding and Heatmap Visualization)

iii). Heatmap Visualization Method:

Real-time Safety Visualization renders locations with color-coded safety indicators:

- Green zones: High safety scores (safer areas)
- Red zones: Low safety scores (higher risk areas)
- 500-meter radius: Localized safety assessment for route segments

iv.) Emergency Features Integration:

- SOS Alert: 98% reliability with 2.3s average response time
- Real-time Location Sharing: 5-10 meter accuracy via GPS coordinates
- Battery Monitoring: Automatic alerts at 15% battery threshold

Algorithm 2: Safety Score Visualization

```
function prepareHeatmapCircles(points)
    circles = {}
    for each point in points:
        intensity = 1 - point.safetyScore
        color = lerpColor(green, red, intensity)
        circles.add(Circle(center=point,
        radius=300, fillColor=color))
    store circles in heatmapCircles
end function
function renderMap():
    mapCircles = isHeatmapVisible ?
    heatmapCircles : {}
    displayMap(circles=mapCircles)
end function
```

This comprehensive overview demonstrates thorough understanding of the FEARLESS system's technical implementation, mathematical foundations, and practical applications for women's safety enhancement through intelligent route optimization and emergency response mechanisms.

Limitations

- **Hands-Free SOS:** The system lacks a voice-activated SOS trigger for emergency situations where manual operation isn't possible.
- **Nearby Volunteer Network:** There is no integrated volunteer network to provide immediate local assistance during emergencies.
- **Periodic Safety Check:** The application does not include an automatic periodic check-in feature to ensure the user's continuous safety.
- **Offline Availability:** The application heavily depends upon internet connectivity and cannot function effectively in low-network or offline conditions.
- **Limited dataset:** dataset and algorithm are presently restricted to Pune city; scalability and performance in other cities remain untested.
- **Outdated model:** This model is not more advanced to accurately predict emerging risk areas and incorporate continuous user feedback for algorithm improvement.

Our Solution: SafeHer

- An integrated, AI-powered web platform designed to be a reliable safety companion.
- SafeHer allows users to send instant alerts, share their live location, and automatically capture evidence, all triggered by a single click or a simple voice command and it also allows to do a fake call to avoid unwanted dangerous circumstances.

Our main purpose is to leverage modern web technologies to create an accessible and intelligent safety net.

Core Features of Our Prototype

We have successfully implemented a prototype with five very useful, interconnected features:

1. **Emergency SOS System:** A single - click and voice-activated alert system.

2. **Voice Command Feature:** Hands-free "Help Me" voice trigger for discreet activation.

3. **Auto Evidence Capture:** Instantly captures and uploads a photo from the webcam upon activation.

4. **Live Location Sharing:** Continuously tracks and shares the user's location in real-time on a map.

5. **Fake Call Activation:** Activate fake call in few seconds from added contacts to avoid a potential danger.

Technology Stack

Component	Technology Used	Purpose
Frontend (UI)	React.js, HTML5, CSS3	Building a responsive user interface.
Backend (API)	Node.js, Express.js	Handling alerts and business logic.
Database	Firebase Realtime Database	Storing & syncing live location data.
Cloud Storage	Firebase Storage	Securely storing captured evidence.
Alerts (SMS)	Twilio API	Sending SMS notifications.
Voice command	Web Speech API	Recognizing the "Help Me" command.
Mapping	Google Maps Platform	Displaying the live tracking map.
Collaboration	Git & GitHub	Version Control and teamwork.

Deep Dive: SOS & Evidence Capture

How it Works :

Trigger 1

The user either clicks the large "SOS" button or says the phrase "Help me."

Instant Action 2

The system immediately sends an SMS alert via the Twilio API to pre-defined emergency contacts.

Evidence 3

Simultaneously, it accesses the device's webcam, captures a photo, and securely uploads it to Firebase Storage, creating a timestamped piece of evidence.

Impact: This provides an immediate alert and preserves crucial information from the moment the emergency begins.

Deep Dive: Live Location Sharing

How it Works :

Activation 1

Once the SOS is triggered, the app begins continuously tracking the device's GPS coordinates using watchPosition.

Real-time Sync 2

Every few seconds, the new coordinates are sent directly to Firebase Realtime Database.

Tracking 3

A unique, shareable link leads to a tracking page where contacts can see a marker moving on a Google Map in real-time.

Impact: This provides loved ones with the user's live, up-to-the-second location, which is critical for a fast response.

Deep Dive: Fake Call Activation

Trigger 1

Instant Action 2

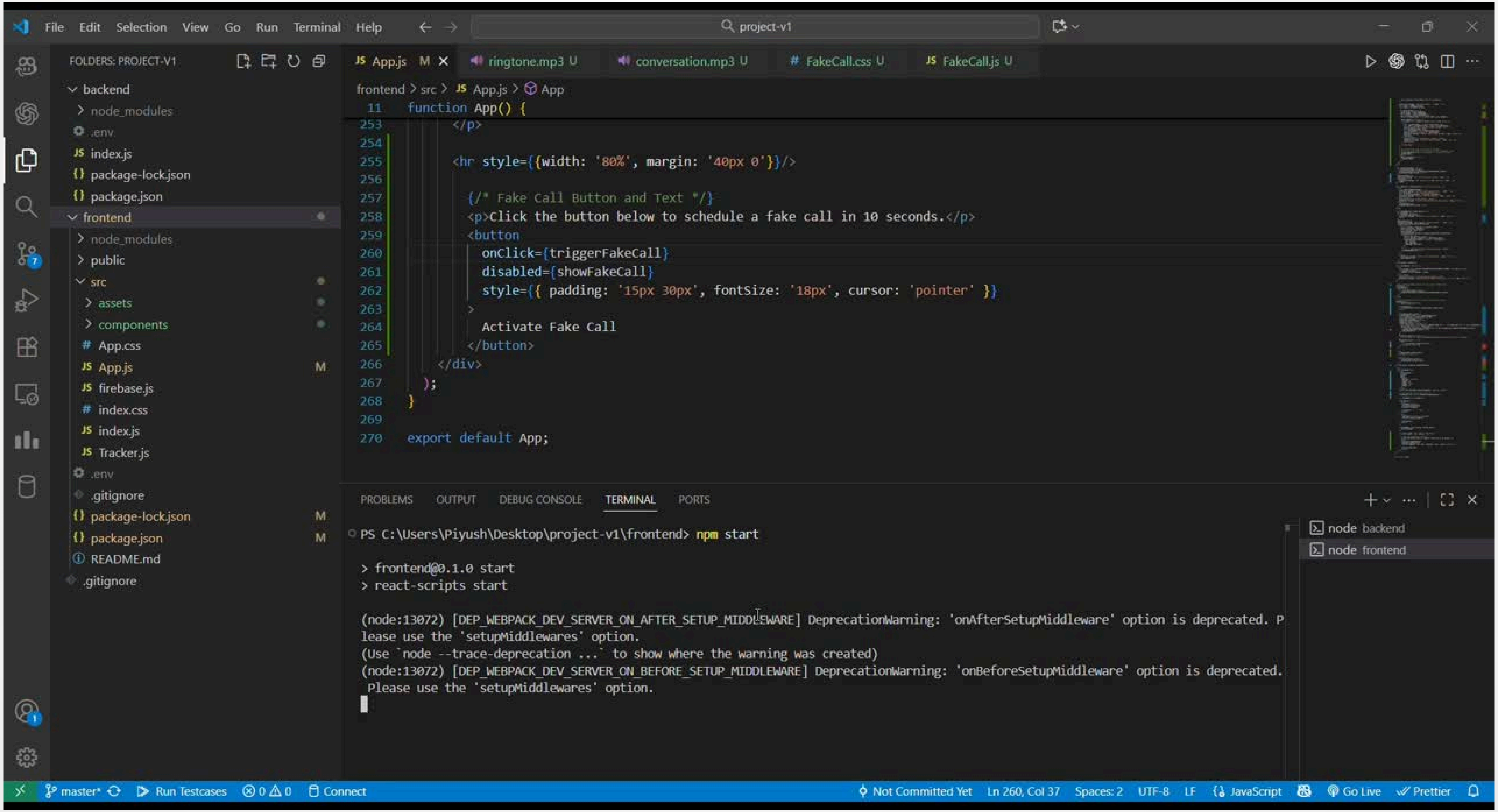
Realistic Interaction 3

How it Works :

1. The user taps the "Fake Call" button, uses a shortcut key.
2. The system schedules and displays a simulated incoming call screen with ringtone, vibration, and caller details to appear real.
3. When answered, a pre-recorded or AI-generated voice plays a short conversation, helping the user safely exit the uncomfortable situation.

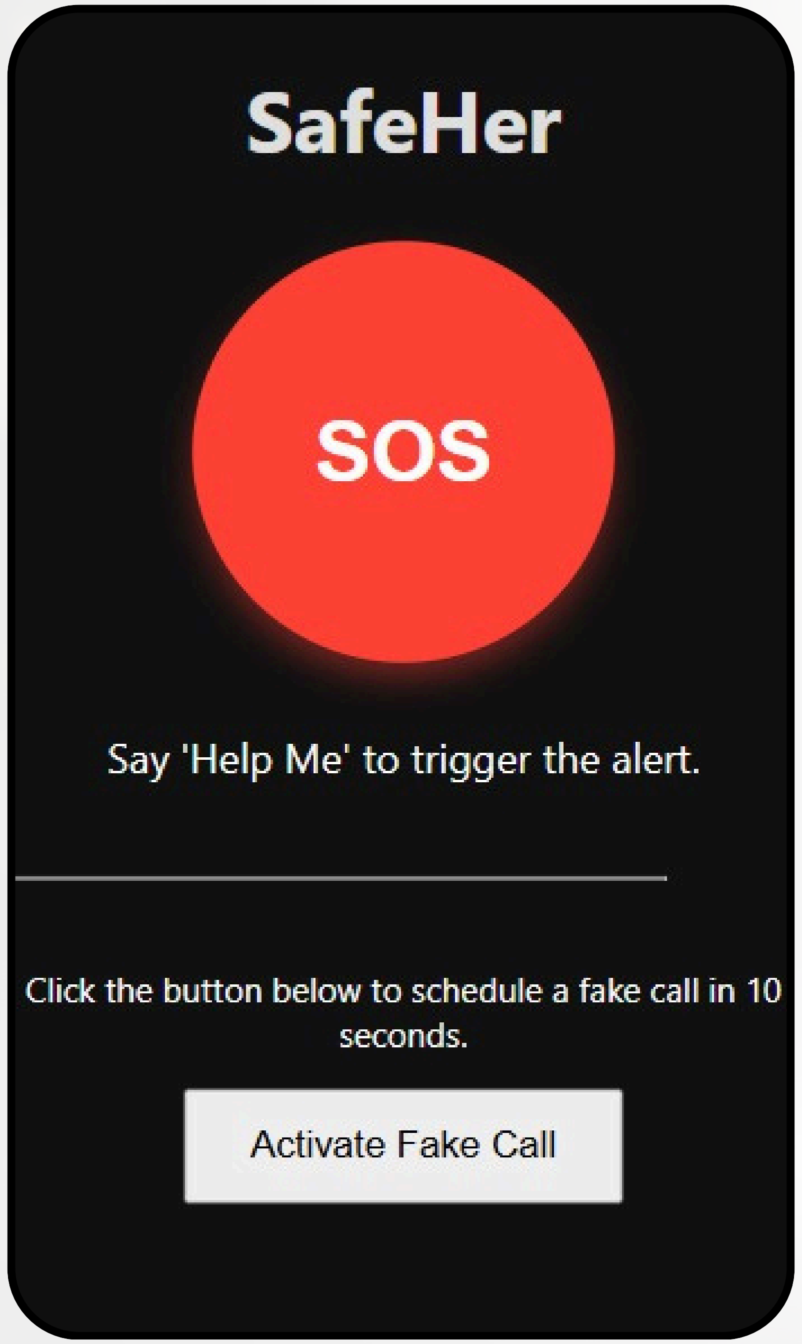
Impact: Empowers users to discreetly take control of a potentially dangerous situation,

System Workflow

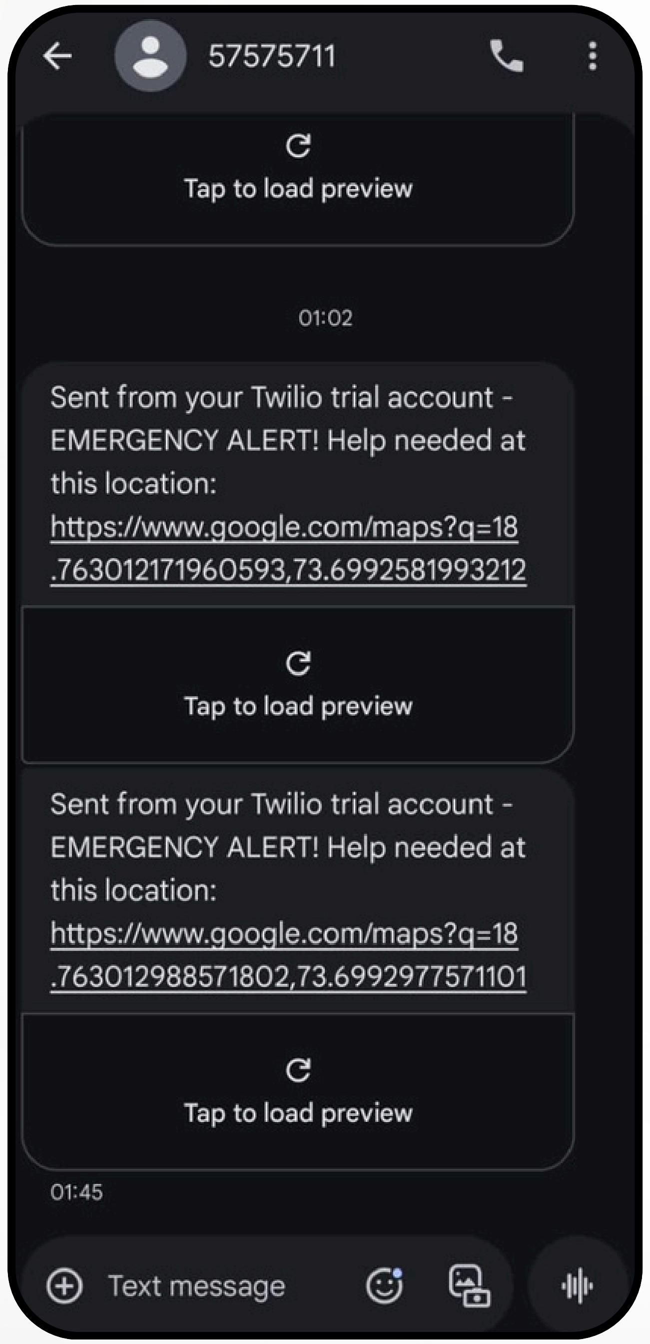


Prototype Visuals

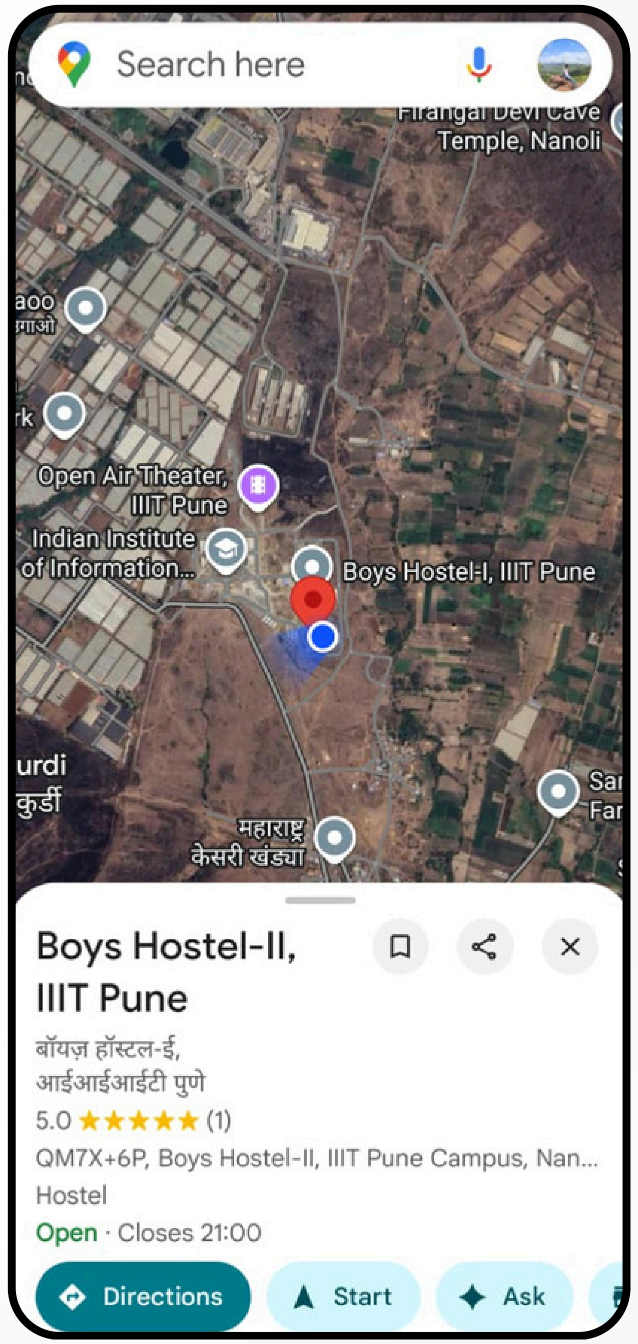
Landing Page



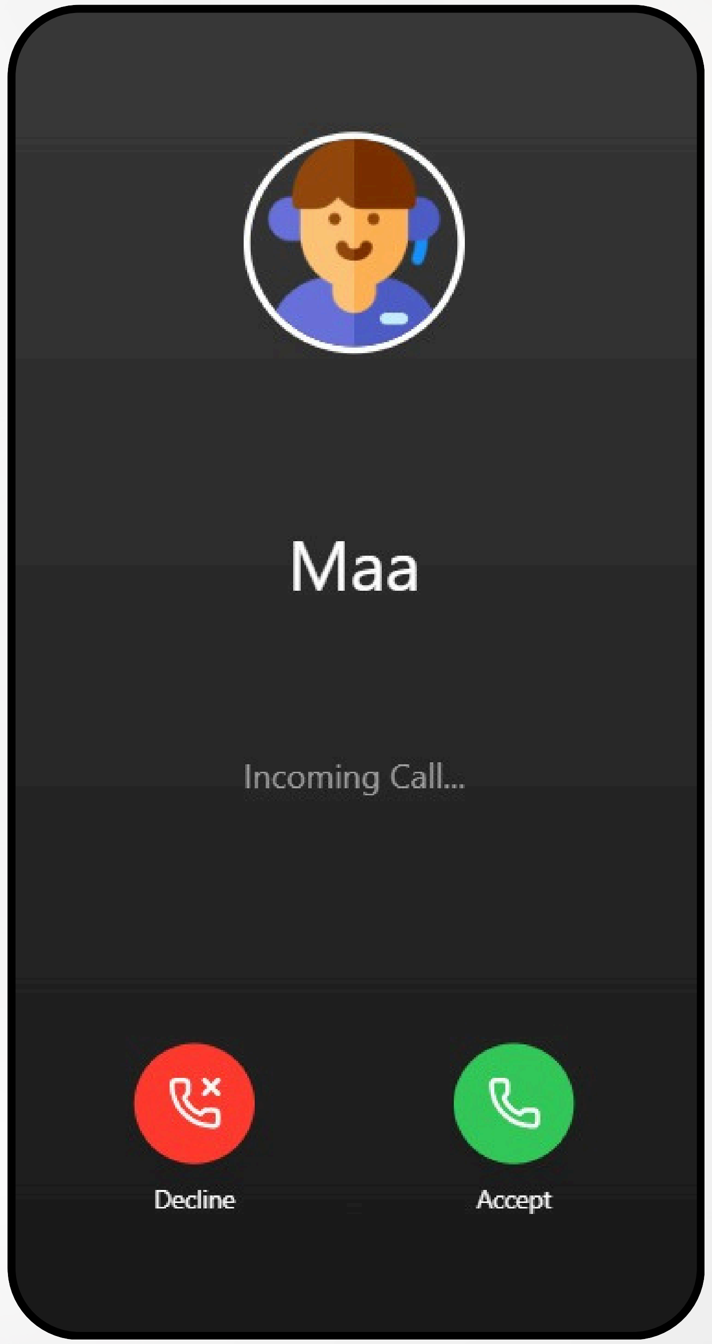
SMS



Location



Fake Call



New Core Features of Our Prototype

We have successfully implemented five interconnected new features in our prototype:

1. **Secure User Authentication:** Personalized accounts with secure login through email and dedicated profiles to store emergency contacts and user roles.
2. **Nearby Volunteer Network:** A "Go On-Duty" system for trusted volunteers to broadcast their location and receive alerts for nearby emergencies.
3. **AI Route Safety Score:** A data-driven tool using XGBoost ML to analyze crime stats and predict safety ratings for different areas in Pune.
4. **AI Safety Chatbot:** An integrated chatbot powered that provides immediate, logic-driven advice and de-escalation steps for unprecedented safety concerns.
5. **Periodic Safety Check:** A proactive timer that automatically triggers an SOS alert if the user does not mark themselves as safe within a set duration, ensuring protection even when incapacitated.

Deep Dive: User Authentication

How it Works :

Secure Authentication

1. Implemented a robust login/signup system using Firebase Auth to verify user identities and protect personal data.

Personalized Profile

2. Each user has a dedicated profile that securely stores critical information like emergency contact numbers.

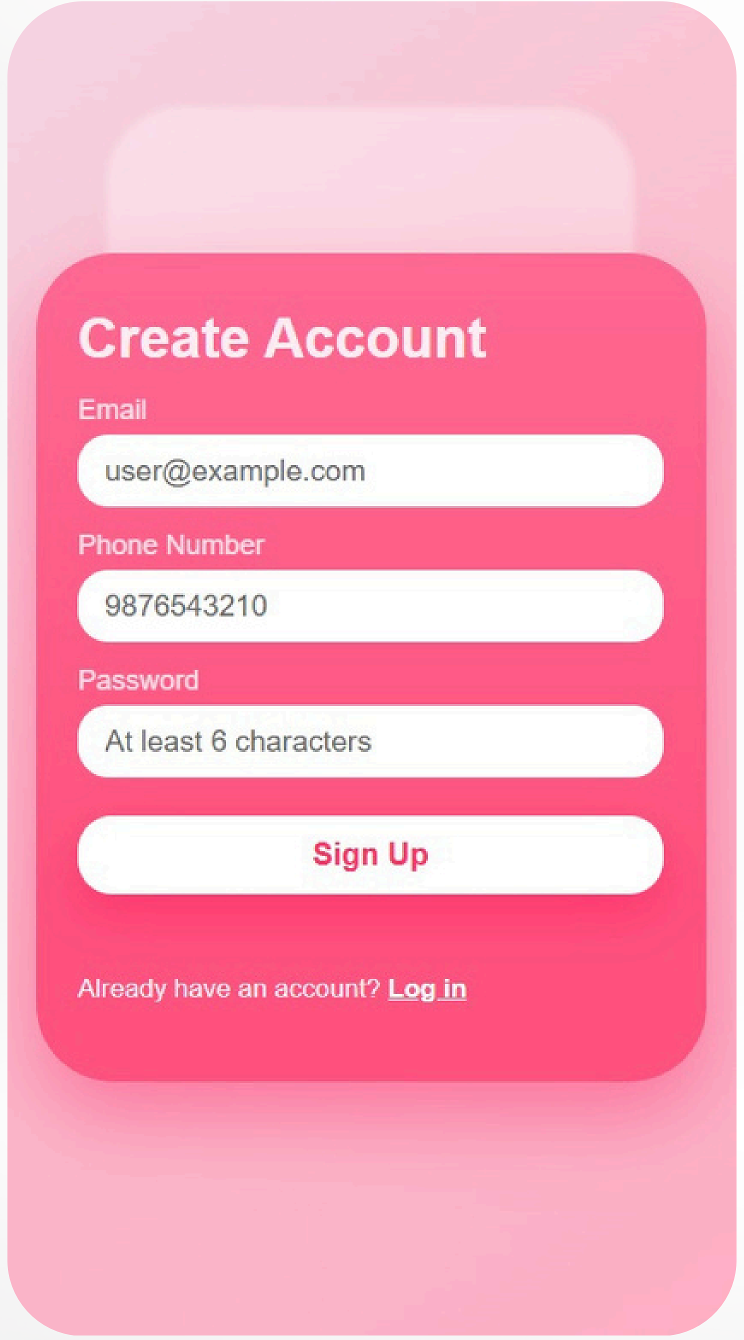
Role-Based Access

3. The system intelligently distinguishes between "Users" (who need help) and "Volunteers" (who provide help).

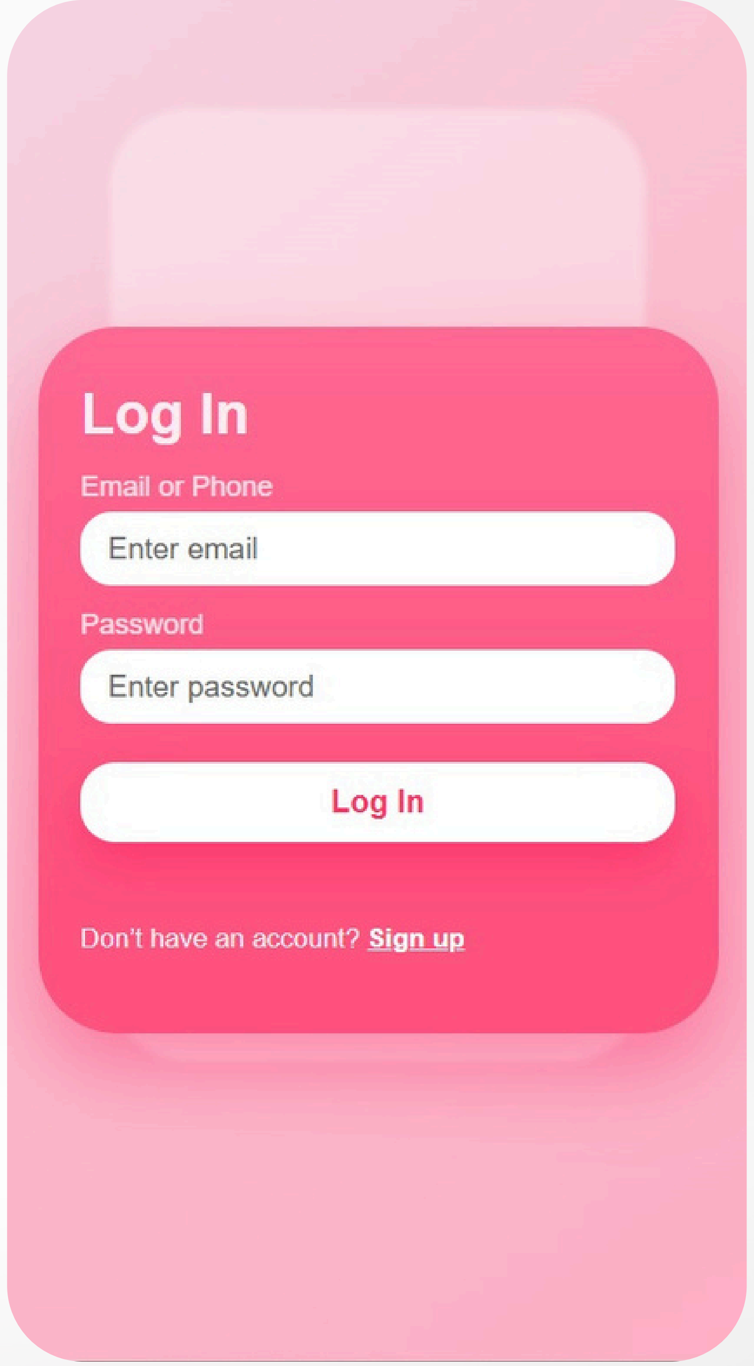
Impact: Establishes a trusted, secure environment where user identities are verified and critical emergency contacts are instantly accessible.

Authentication Visuals

SignUp Page



Login Page



Deep Dive: Volunteer Network

How it Works :

The "On-Duty" System

1 Trusted volunteers can toggle an "On-Duty" mode, broadcasting their live location to our secure geospatial database.

Smart Dispatch Algorithm

2 When an SOS is triggered, the backend automatically calculates the distance to all active volunteers in real-time.

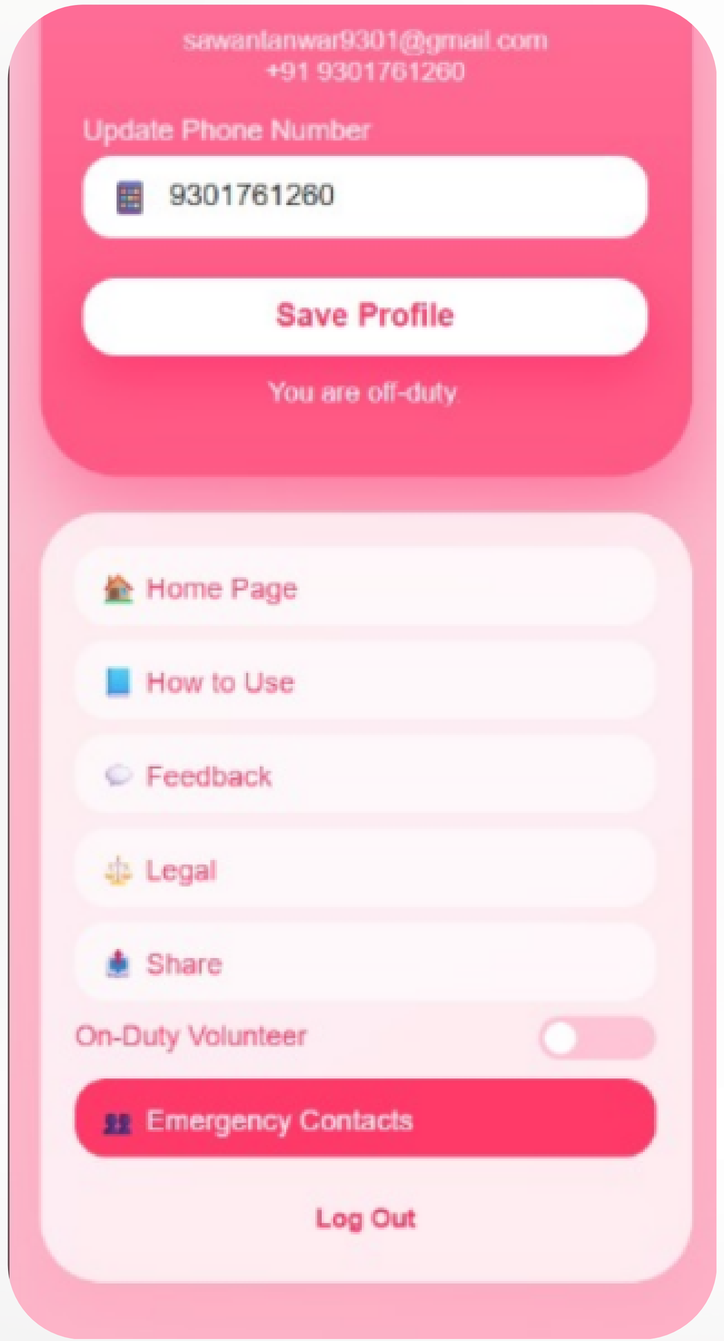
Instant Volunteer Alerts

3 The system identifies the closest volunteers (within a dynamic radius) and sends them an immediate SMS with the victim's location.

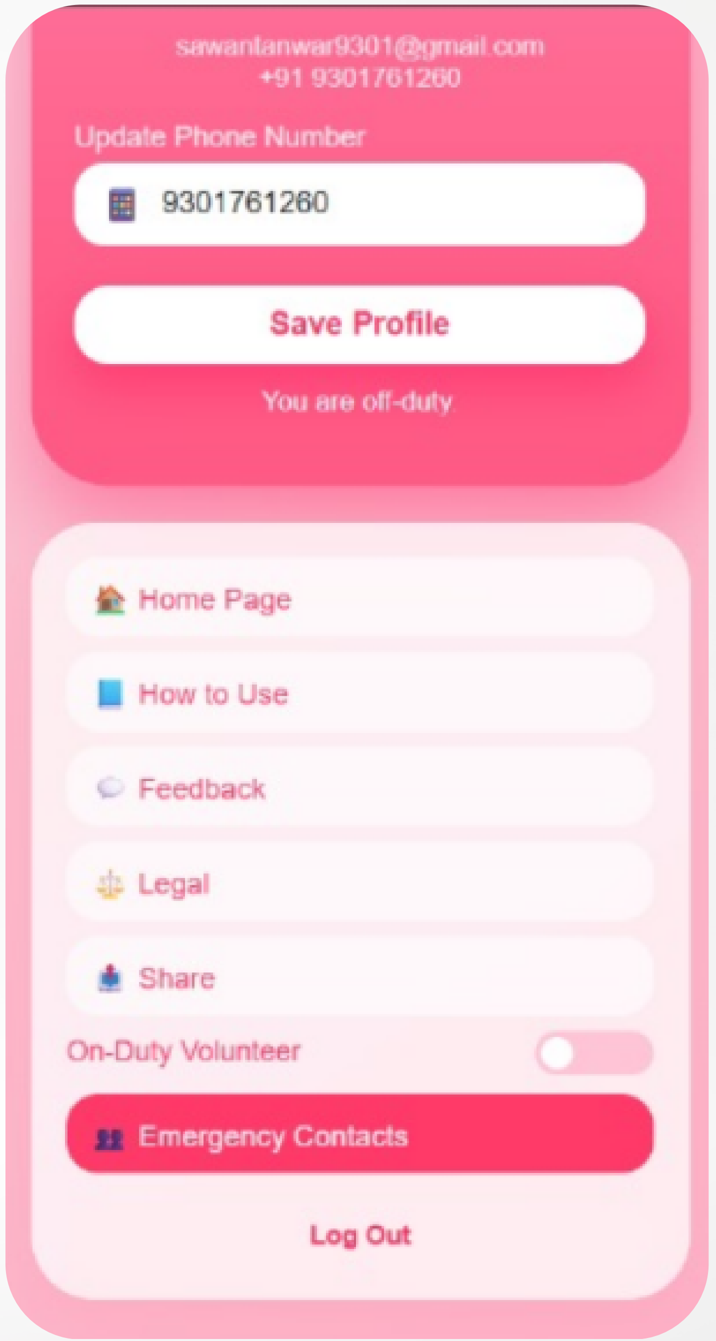
Impact: Creates a hyper-local safety net that drastically reduces response time by crowdsourcing immediate help from trusted individuals nearby.

Volunteer Visuals

Add Volunteer



On/Off Duty



Deep Dive: AI Safety Score

How it Works :

Data-Driven Assessment

1. An intelligent tool that analyzes historical crime data for specific areas in Pune to calculate a real-time "Safety Score" (0-10).

XGBoost Machine Learning

2. Utilizes an advanced Gradient Boosting model to identify complex patterns between crime rates and safety levels.

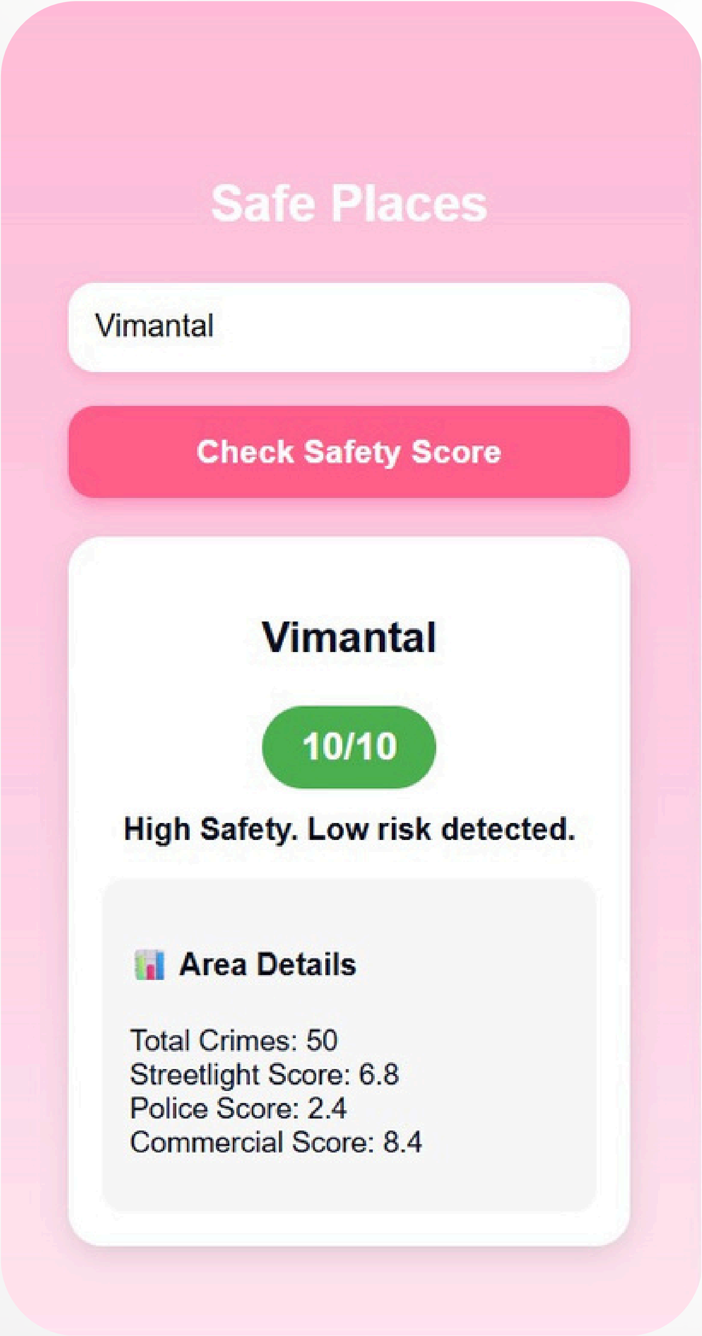
Proactive Prevention

3. Empowers users to check the safety rating of a destination before they travel, allowing for safer route planning.

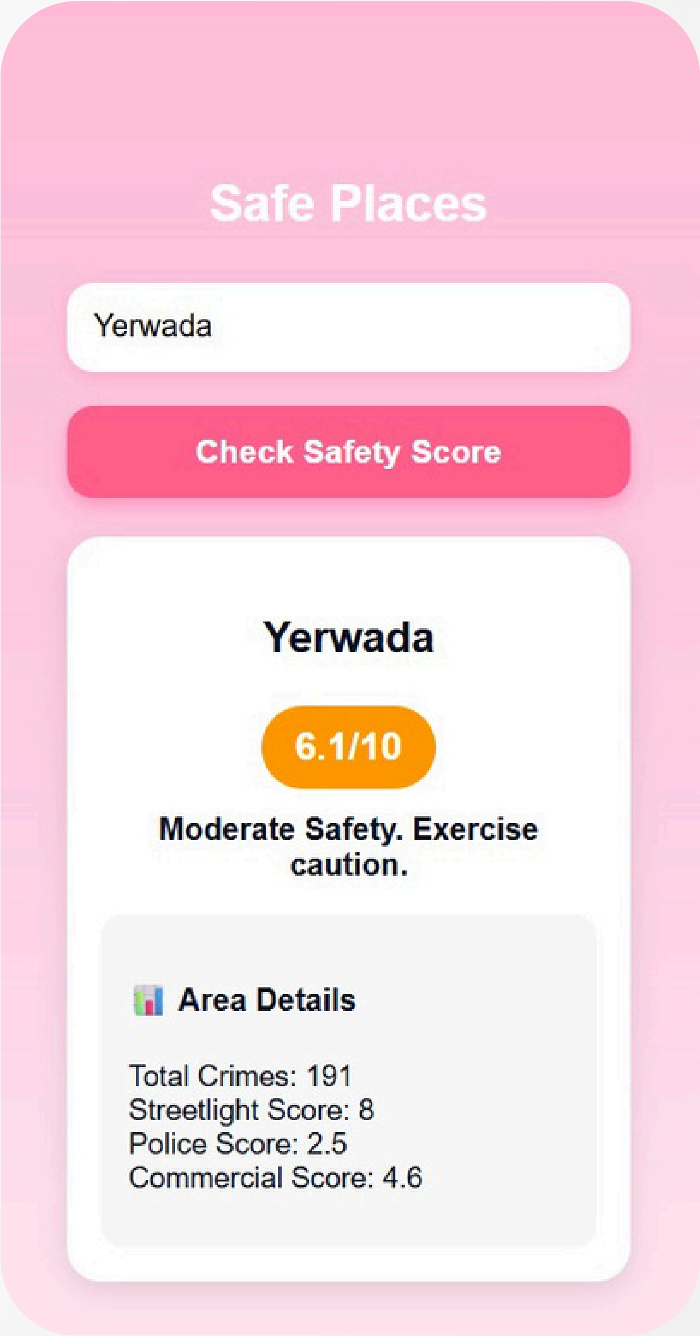
Impact: Shifts the platform from reactive to proactive, empowering users to avoid high-risk areas by making data-driven travel decisions.

Safety Score Visuals

Check Safety Score



Check Safety Score



Deep Dive: Periodic Safety Check

How it Works :

Proactive Monitoring

1. A user-initiated timer that prompts for safety confirmation at regular intervals (e.g., every 10 minutes) during a journey.

Fail-Safe Trigger

2. If the user fails to respond or marks themselves as unsafe, the system automatically escalates to a full SOS alert.

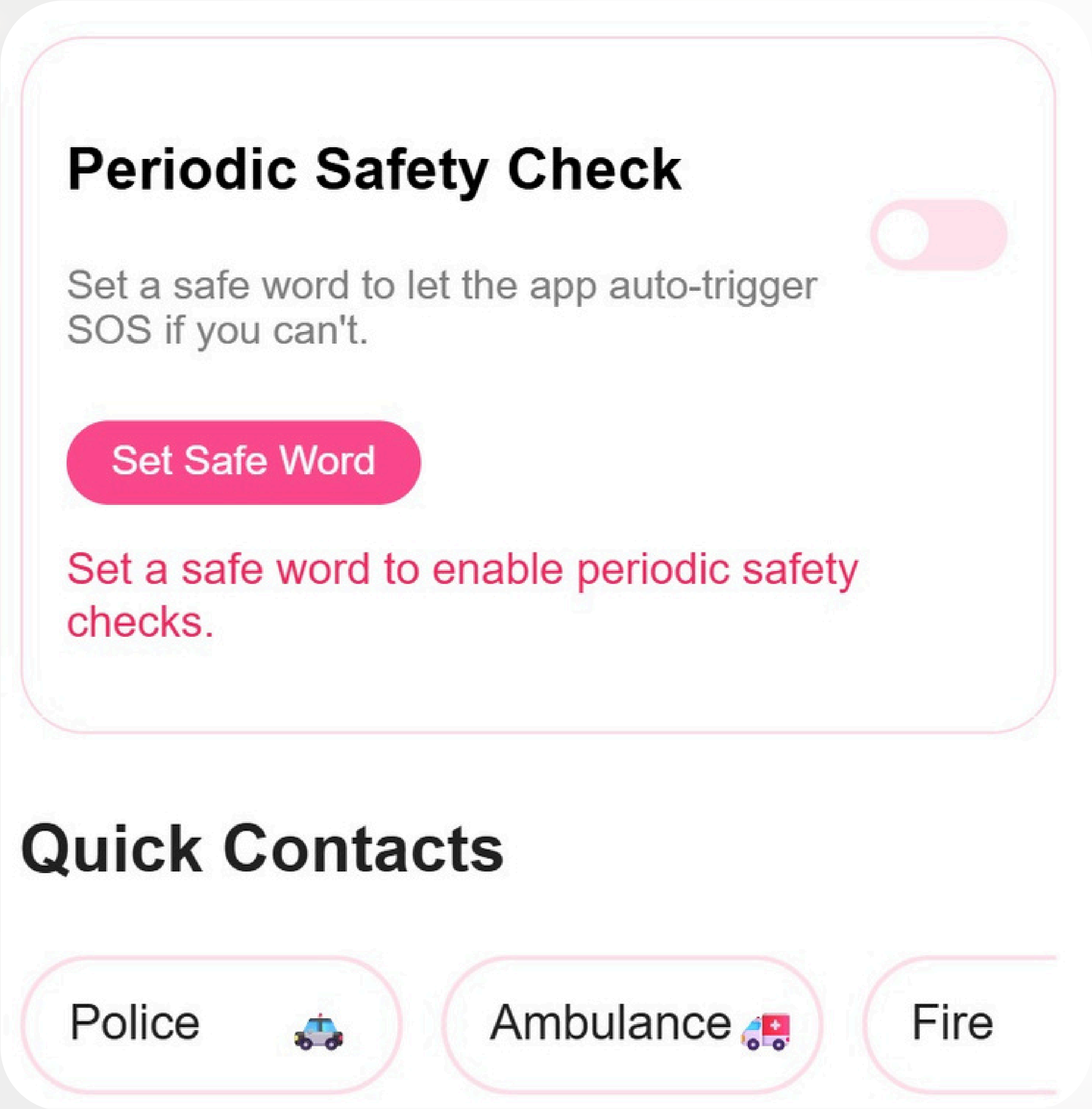
Continuous Protection

3. Ensures that help is summoned even if the user is incapacitated and unable to manually trigger an alarm.

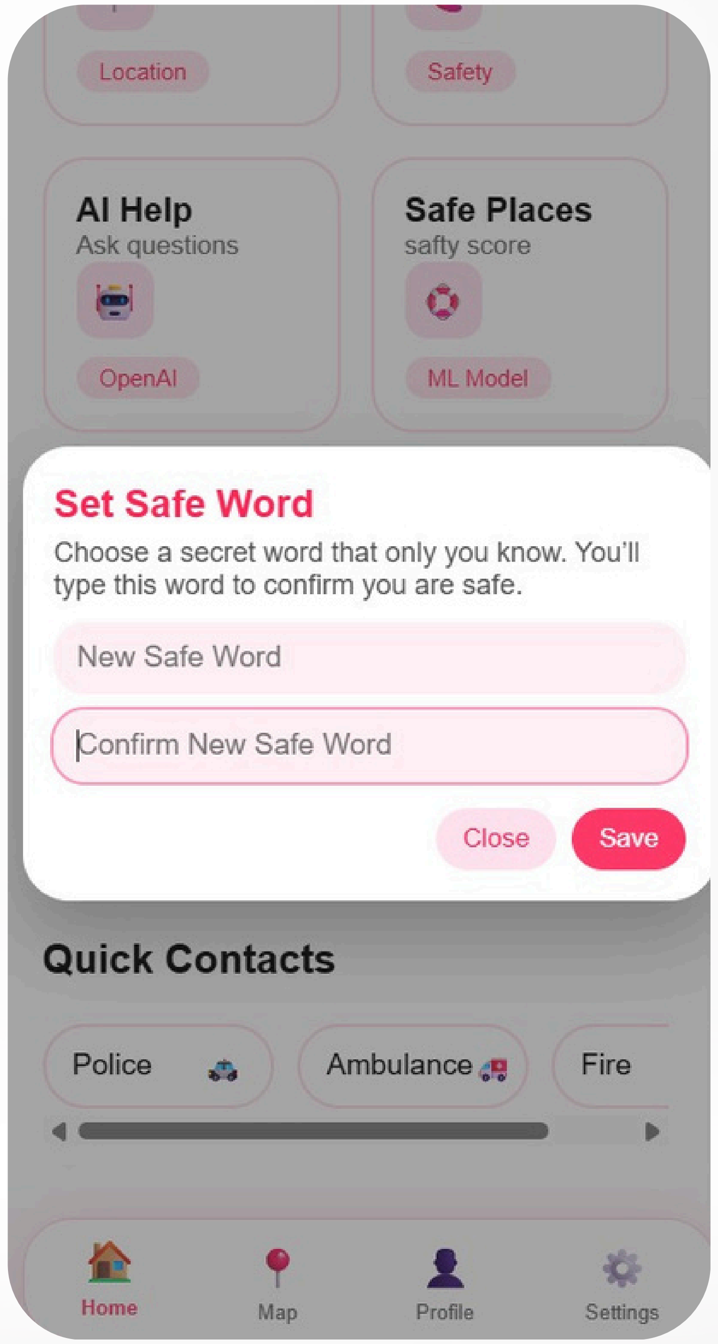
Impact: Provides a safety net for solo travelers by automating the distress signal if they become unable to call for help themselves.

Periodic Safety Visuals

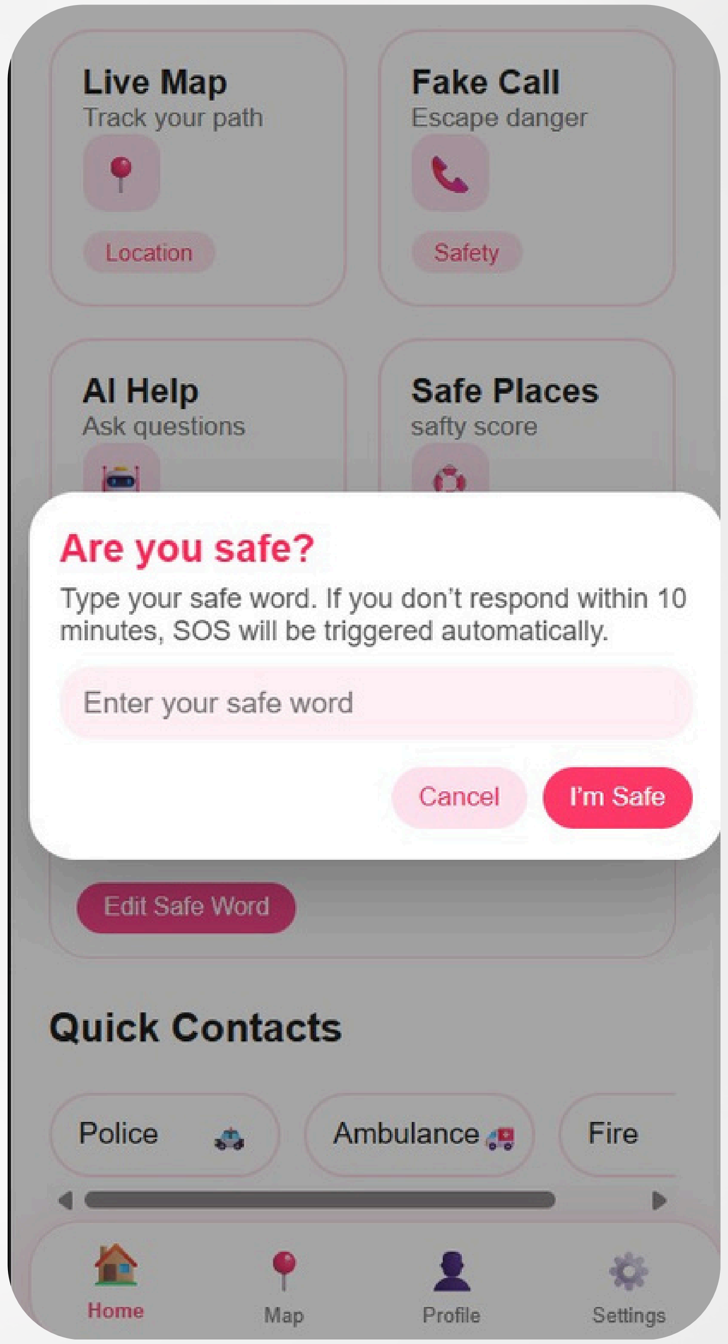
Periodic Safety Check



Set Safety Word



Check Safety



Deep Dive: AI Safety ChatBot

Instant Guidance

Context-Aware Assistance

One-Tap Access

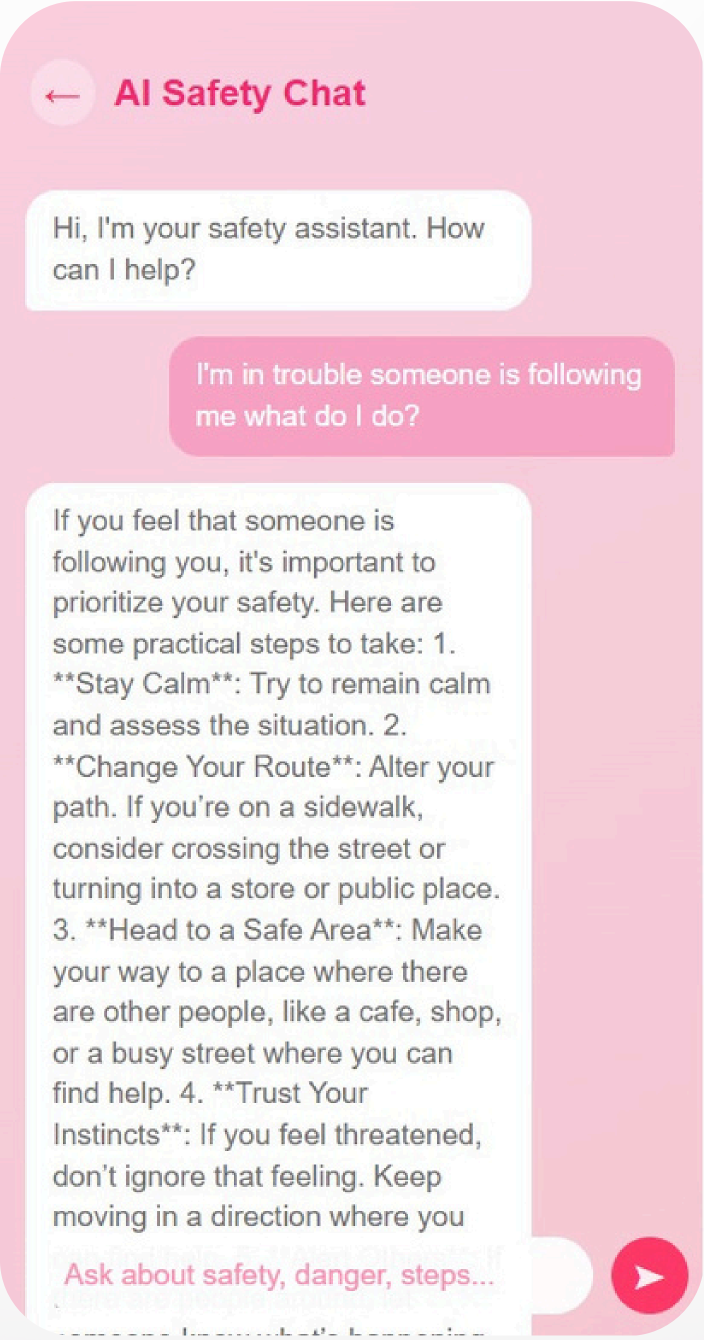
How it Works :

1. An integrated AI chatbot powered by the OpenRouter API that provides immediate advice for unprecedented safety concerns.
2. Users can ask specific questions about their situation (e.g., "I'm being followed") and receive calm, actionable steps to de-escalate or find safety.
3. Easily accessible via a dedicated button in the UI, allowing users to quickly seek help without navigating away from the safety dashboard.

Impact: Empowering users with an intelligent companion that offers real-time, logic-driven solutions for complex or confusing safety scenarios.

AI Safety ChatBot Visuals

AI ChatBot



Future Scope

1. SafeHer App

Since the current prototype operates as a web-based system with limited background capabilities, future development will focus on transforming SafeHer into a fully secure Android and iOS mobile application to enable deep sensor integration, improved offline handling, and enhanced emergency response mechanisms.

2. Offline Availability

Future enhancements will allow the application to store essential safety maps, crime clusters, volunteer data, and safety scores locally, ensuring uninterrupted access to critical features even in areas with weak or no internet connectivity.

3. AI-Based Voice Distress Detection

An advanced voice-analysis model can be integrated to detect panic or distress signals in the user’s voice, allowing automatic SOS activation even when predefined commands cannot be spoken due to fear or physical constraints.

Future Scope

4. Continuous Model Improvements

The Safety Score prediction model can be updated dynamically using user feedback, incident confirmations, and volunteer activity data, enabling continuous retraining and improved accuracy as crime patterns evolve.

5. Integration with Government & Public Safety Systems

Future versions may connect directly with police APIs, municipal surveillance systems, and public safety alerts, enabling faster emergency escalation and access to official safety data.

6. Wearable & IoT Compatibility

SafeHer can be extended to function with smartwatches, fitness bands, and IoT devices, enabling discreet SOS triggers such as double-tap gestures, squeeze inputs, or heart-rate anomaly detection.

7. Advanced Community-Driven Safety Features

Upcoming enhancements may include neighborhood safety dashboards, real-time volunteer heatmaps, and AI-optimized safe route recommendations to enhance preventive safety and community engagement.

References

1. P. Premi, K. S. Savita and N. Millatina, "FRNDY: A Women's Safety App," 2022 6th International Conference On Computing, Communication, Control And Automation (ICCUBEA, Pune, India, 2022, pp. 1-5, doi: 10.1109/ICCUBEA54992.2022.10010815. keywords: {Uniform resource locators;Legged locomotion;Passwords;Software;Hardware;Safety;Pins;Women safety;Android application;Smartphone},

2. Mandapati, S., Pamidi, S. and Ambati, S., 2015. A mobile based women safety application (I Safe Apps). IOSR Journal of Computer Engineering (IOSR-JCE), 17(1), pp.29-34.

3. D. Chand, S. Nayak, K. S. Bhat, S. Parikh, Y. Singh and A. A. Kamath, "A mobile application for Women's Safety: WoSApp," TENCON 2015 - 2015 IEEE Region 10 Conference, Macao, China, 2015, pp. 1-5, doi: 10.1109/TENCON.2015.7373171. keywords: {Law enforcement;Mobile applications;Safety;User interfaces;Databases;Google;Global Positioning System;Humanitarian Technology;Women's Safety;Smartphone Application},

4. A. S. Ummer, B. Mariam Biju, J. M. Noble, S. K U and D. K S, "HER SHIELD: Speech Emotion Recognition for Women's Safety," 2025 2nd International Conference on Trends in Engineering Systems and Technologies (ICTEST), Ernakulam, India, 2025, pp. 1-6, doi: 10.1109/ICTEST64710.2025.11042720. keywords: {Emotion recognition;Sentiment analysis;Geology;Speech enhancement;Logic gates;Market research;Robustness;Real-time systems;Mobile computing;Speech to text;Speech Emotion Recognition;Natural Language Processing;Convolutional Neural Networks;Women's Safety;Mobile Computing},

5. K. Ghuge, S. Khillari, A. Sadawrate, R. Varpe, A. Shaikh and R. Desai, "FEARLESS: Women Safety Software," 2025 International Conference on Computing Technologies (ICOCT), Bengaluru, India, 2025, pp. 1-6, doi: 10.1109/ICOCT64433.2025.11118864. keywords: {Navigation;Urban areas;Pressing;User interfaces;Real-time systems;Software;Safety;Mobile applications;Planning;Testing;Safe Route Navigation;Emergency Alert;Real-Time Tracking;Mobile Application;Women's Safety},

References

6. Abhaya, V., & Sheela, T. (2017). Smart walking stick for visually impaired and women safety. 2017 International Conference on Innovations in Information, Embedded and Communication Systems (ICIIECS), 1-4. https://doi.org/10.1109/ICIIECS.2017.8276143

7. Chaudhari, R., & Pawar, S. (2019). Women safety system using IoT. 2019 3rd International Conference on Trends in Electronics and Informatics (ICOEI), 747-750. https://doi.org/10.1109/ICOEI.2019.8862662

8. Mahajan, S., & Dhiwar, S. (2020). Intelligent safety system for women security. International Journal of Engineering Research & Technology (IJERT), 9(5), 526-529.

9. Pathan, N. S., & Jaiswal, S. (2021). A machine learning approach for safe route prediction. 2021 International Conference on Artificial Intelligence and Smart Systems (ICAIS), 154-159. https://doi.org/10.1109/ICAIS50930.2021.9395921

10. Reddy, P. K., & Reddy, G. R. (2018). Women safety application using GPS and GSM. International Journal of Computer Science and Mobile Computing, 7(4), 145-150.

11. Sharma, A., & Singh, B. (2022). Android based women safety application with live location tracking. 2022 10th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO), 1-5. https://doi.org/10.1109/ICRITO56286.2022.9964871

THANK YOU

FOR YOUR TIME & PATIENCE